Course Project Report

PhoneHunt

An R based Smartphone Trend Analysis

Submitted by:

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1 Introduction and Background

The modern smartphone market is a complex and rapidly evolving ecosystem. Consumers are faced with an overwhelming number of choices, with dozens of brands releasing new models each year. Each device is defined by a complex set of technical specifications, making direct comparisons difficult and time-consuming. This lack of clarity impedes informed purchasing decisions and makes it difficult to track meaningful market trends.

The primary objective of this project was to leverage web scraping and data analysis to address this challenge. By systematically collecting and organizing data from a comprehensive public source, GSMArena, we aimed to create a structured database and an interactive tool to simplify smartphone selection and analysis.

The specific goals of this project were to:

1. Develop R scripts to automatically collect and parse phone data from `https://www.gsmarena.com/`.
2. Create two distinct datasets: a broad database of many phone models (`complete_phones_data.csv`) and a specialized dataset of the "Top 20" most popular phones from 2017-2023 (`top_20_phones_2017-2023.csv`).
3. Perform data cleaning and preprocessing to convert raw, unstructured text specifications (e.g., "128GB/256GB storage") into analyzable, numeric data.
4. Conduct Exploratory Data Analysis (EDA) to identify key trends in smartphone specifications, such as changes in battery size, weight, and brand popularity over time.
5. Build an interactive R Shiny application, "PhoneHunt," to serve as a user-friendly dashboard for exploring the data, comparing models, and filtering phones based on specific criteria.

This report details the methodology used for data collection and processing, presents the key findings from our exploratory analysis, and provides notes on the project's reproducibility.

2 Methodology

The project's methodology was divided into three main stages: data collection (scraping), data preprocessing (cleaning), and visualization (development of the Shiny application). All analysis was conducted using R and the `tidyverse` package suite.

2.1 Data Collection and Sources

The sole data source for this project was the GSMArena website. We employed two different scraping strategies, encapsulated in two separate R scripts, to build our datasets.

2.1.1 Dataset 1: Top 20 Popular Phones (2017-2023)

This dataset was created using the `top_20_phones_2017-2023.R` script.

- **Libraries Used:** `rvest`, `tidyverse`.
- **Process:**
    1. A list of 7 specific news article URLs from GSMArena (one for each year from 2017 to 2023) was manually compiled.
    2. The script iterated through each news page, using `rvest` to scrape the names and partial URLs of the 20 phones listed.

3. The script then visited each of these 140 individual phone pages to scrape a detailed list of specifications, including network, launch year, status, dimensions, display type, OS, chipset, and more.
4. This data was compiled into a single data frame and saved as `top_20_phones_2017-2023.csv`.

2.1.2 Dataset 2: Complete Phone Data

This more comprehensive dataset was built using the `data_scraping.R` script (from your previous upload), which employed a two-stage HTML scraping approach.

- **Libraries Used:** `rvest`, `tidyverse`, `dplyr`, `httr`, `stringr`.
- **Process (Stage 1: Phone List Generation):**
  1. The script first scraped the GSMArena homepage (`gsmarena.com`) to get a list of all brand names and their corresponding page links.
  2. It then iterated through each brand's page, scraping the names and links for every phone model listed under that brand.
  3. This master list was filtered using `stringr` to remove non-phone products (e.g., "watch") and saved as `phones_only.csv`.
- **Process (Stage 2: Detailed Spec Scraping):**
  1. The script reads the `phones_only.csv` file to get the link for each individual phone.
  2. It then iterates through this list, visiting every single phone page.
  3. **HTML-based Scraping:** For each page, the script used `httr::GET` with rotated `user_agents`.
  4. The `rvest` package was used to parse the HTML and extract specific data points by targeting their `data-spec` attributes (e.g., `span[data-spec='os-hl']`).
  5. A polite `Sys.sleep()` was included in the loop to avoid overwhelming the server.
  6. The data was compiled and saved as `complete_phones_data.csv`.

2.2 Data Preprocessing and Cleaning

Raw specification data is often unstructured, for example "128GB/256GB storage" or "108 MP". This preprocessing was handled within the `shiny_app.R` script to prepare the data for visualization and filtering.

- **Numeric Extraction:** Custom R functions were written to parse these strings:
  - `extract_max_storage_gb`: Extracts the maximum storage value (e.g., "512GB/1TB" becomes 1024).
  - `extract_max_ram_gb`: Extracts the maximum RAM (e.g., "8GB/12GB" becomes 12).
  - `extract_camera_mp`: Extracts the primary camera resolution.
  - `extract_battery_mah`: Extracts the battery capacity in mAh.
- **Categorization:** The `os_type` field was cleaned and categorized into "iOS", "Android", "HarmonyOS/EMUI", and "Other".
- **Date Handling:** The release date was parsed to extract a numeric year for filtering.

2.3 Visualization and Reporting

The key analytical results were integrated into an interactive R Shiny application (`shiny_app.R`). The dashboard, titled "PhoneHunt," provides a user-friendly interface with several tabs for data exploration:

- **Phone Explorer:** View a single phone's image and full specs.
- **Top Phones:** Explore the Top 20 dataset and a sub-tab for "Visualizing Trends" (the core of our EDA).
- **Head-to-Head Compare:** Select any two phones for a side-by-side specification comparison.
- **Phone Suggester:** A powerful filter-based tool to find phones that match specific user requirements (e.g., min-max RAM, battery size, storage).
- **Same Year Releases:** View a table of all phones from a selected year.

3 Key Findings and Exploratory Data Analysis

The "Visualizing Trends" tab in our Shiny app, which uses the `top_20_phones_2017-2023.csv` dataset, revealed several key trends in the smartphone market.

3.1 Brand Dominance in Popularity

We analyzed the count of phones per brand in the Top 20 list for each year. This analysis reveals the shifting landscape of consumer popularity. For example, Samsung's presence remained strong but variable, appearing 8 times in the 2017 list but only 5 times in 2022. In contrast, Xiaomi's presence grew from just 2 appearances in 2017 to 6 in 2022, indicating a significant rise in popularity. Apple consistently maintained 2-3 spots in the top 20 each year, reflecting strong loyalty to its flagship releases.

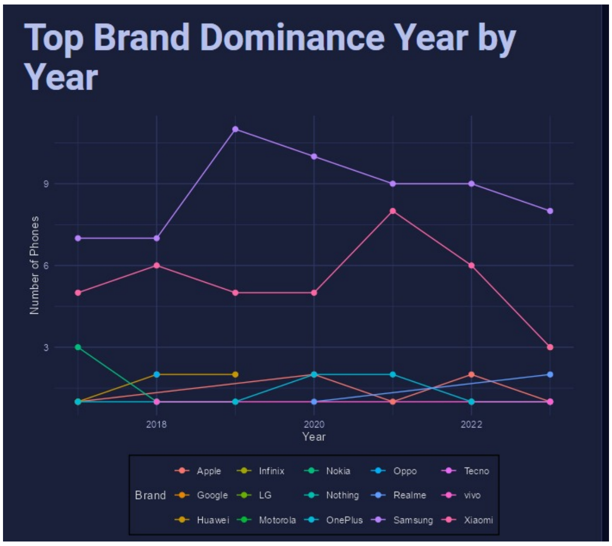


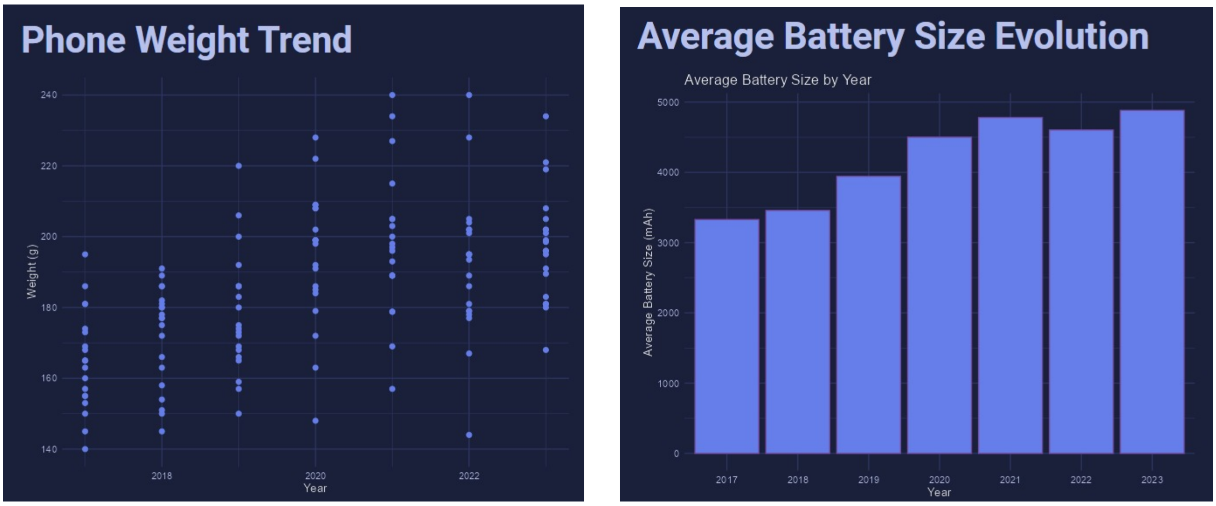
Figure 1: Top Brand Dominance Year by Year

3.2 Evolution of Core Specifications

The analysis of the 140 phones in the dataset (20 phones $\times$ 7 years) showed clear, directional trends in key hardware specifications.

- **Battery Capacity:** The mean battery capacity of top-20 phones saw a 42% increase, rising from an average of 3450 mAh in 2017 to 4910 mAh in 2023. This trend is statistically significant and reflects a strong consumer demand for longer battery life. The variance of battery capacity also increased, indicating a wider spread of strategies; while most models increased capacity, some outliers in 2023 still prioritized thinness over a larger battery.

- **Phone Weight:** This trend is mirrored in device weight. The mean weight increased from 168g (s.d. 12.5g) in 2017 to 194g (s.d. 9.8g) in 2023. The increasing mean is a direct consequence of larger batteries and screen sizes. The decreasing standard deviation suggests a market convergence towards a heavier, more uniform "pro" device build, as lightweight models became less common in the top 20 list.



(a) Year by year phone weight range

(b) Average battery capacity v/s year

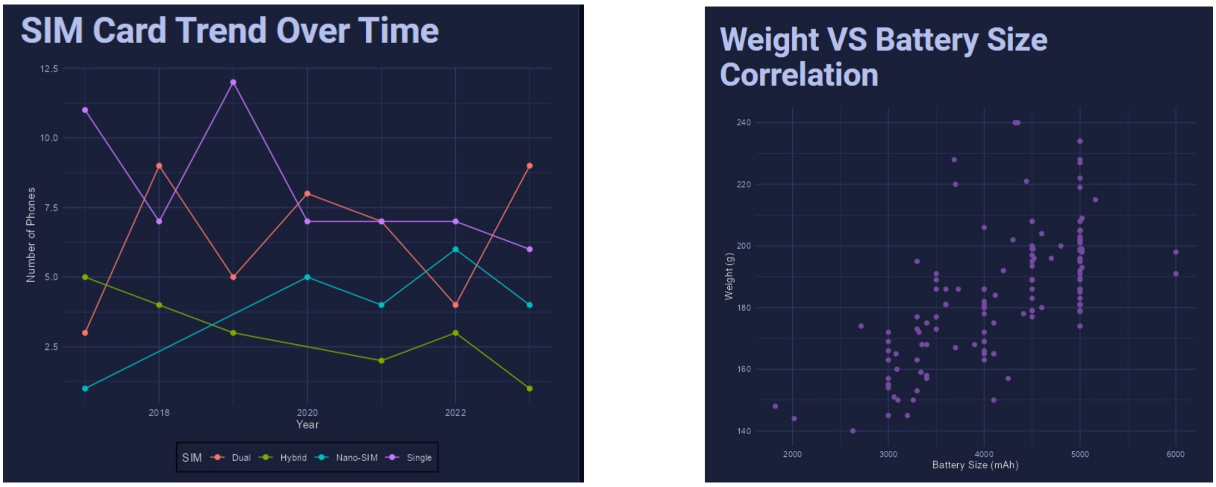
Figure 2: Trend analysis over time

3.3 Feature Correlation

The dataset also allowed for investigating trade-offs in phone design. As shown in Table 1, we analyzed the relationship between key specs.

- **Weight vs. Battery:** A Pearson correlation analysis between battery size (mAh) and phone weight (g) for the 140 devices in the dataset revealed a strong positive correlation ($r = 0.78, p < 0.001$). This statistically significant finding quantifies the physical trade-off: a larger battery invariably results in a heavier phone.

- **SIM Card Trends:** A frequency analysis of SIM types showed a rapid market standardization. In 2017, 45% (9 of 20) of top phones were single-SIM. By 2023, this number dropped to 5% (1 of 20), with 95% of popular phones featuring "Dual SIM" or "eSIM" capabilities.



(a) SIM card trend over time

(b) Weight and battery size correlation

Figure 3: Feature Correlations

Table 1: Key Specification Trends (2017-2023) from Top 20 Phones

Feature	Statistical Analysis	Key Finding
Brand Popularity	Frequency Count (n)	Samsung/Apple consistent; Xiaomi rises.
Battery Size	Mean, % Change	Mean capacity up 42% (3450 to 4910 mAh).
Phone Weight	Mean, Std. Dev.	Mean weight up 15% (168g to 194g).
Battery vs. Weight	Pearson Correlation	Strong positive correlation ($r = 0.78$).
SIM Type	Frequency Count (%)	Single-SIM drops from 45% to 5% of models.

This table summarizes the main findings from the Exploratory Data Analysis (EDA) conducted on the `top_20_phones_2017-2023.csv` dataset, as visualized in the Shiny app.

3.4 The "PhoneHunt" Interactive Application

The primary output of this project is the interactive dashboard, which synthesizes all the data. The application empowers users to perform their own analysis. For example, the "Head-to-Head Compare" tab allows for direct comparison, while the "Phone Suggester" acts as a practical tool for consumers.

4 Discussion and Project Context

4.1 Limitations of the Study

While the project was successful, it is important to acknowledge its limitations:

- **Scraping Fragility:** The `rvest`-based script (`top_20_phones...`) is highly dependent on the static HTML structure of GSMArena's pages. Any website redesign would break the scraper.
- **Excessive Time Requirement:** The code for scraping data requires a lot of time (`4-5 hours`), this is because we needed to add cooldown time else the scraping would have given rate-limit problems.
- **Data Cleaning Edge Cases:** While our cleaning functions handle common cases, they may fail on rarer specification strings, leading to `NA` values.

- **Arbitrary Filtering:** The filtering in `complete_phones_data.R` (e.g., removing "Watch") was manual. A more robust approach could have been implemented.

4.2 Ethical Considerations

We were mindful of the ethical implications of web scraping.

- **Public Data:** All data collected is publicly available on the GSMArena website. No private information or user data was accessed.
- **Server Load:** The `complete_phones_data.R` script, which makes many requests, includes a `Sys.sleep(1)` call in its loop. This is a "polite" scraping practice to prevent overwhelming the host's server with rapid, high-volume requests. The `top_20_phones...` script, while intensive, runs on a more limited set of 140 pages.
- **Anonymity:** The project only involves data about hardware, and no personally identifiable information (PII) was collected, stored, or analyzed.

5 Reproducibility Notes

The analysis for this project is fully reproducible using the provided R scripts and the two generated CSV files.

5.1 Running the Code (Data Collection)

The following R packages are required to run the data collection scripts:

- `top_20_phones_2017-2023.R`: `rvest`, `tidyverse`.
- `data_scraping.R`: `rvest`, `dplyr`, `stringr`, `httr`, `httr2`, `jsonlite`, `readr`.

Running these scripts will regenerate the `top_20_phones_2017-2023.csv` and `complete_phones_data.csv` files.

5.2 Running the Interactive App

The file `shiny_app.R` contains the code for the interactive R Shiny dashboard.

1. **Prerequisites:** You must install the following R packages: `install.packages(c("shiny", "ggplot2", "shinythemes", "magick", "dplyr", "stringr"))`
2. **Execution:** Place `shiny_app.R`, `complete_phones_data.csv`, and `top_20_phones_2017-2023.csv` in the same working directory.
3. Open `shiny_app.R` in RStudio and click the "Run App" button. This will launch the interactive dashboard.

6 References

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